# Expressivity and Penetrance

## Part 1

1) Split your group into two.

2) Each group decide if you will be 'Expressivity' or 'Penetrance.' DO NOT TELL THE OTHER GROUP. Both groups may end up being the same. That's ok.

3) Each group:
   1) Create a scenario that describes and illustrates <u>incomplete</u> Expressivity OR Penetrance (depending on what you decided in '2' above). (you are welcome to use a real phenotype if you would like)
   2) Create a scenario that describes and illustrates <u>complete</u> Expressivity OR Penetrance (depending on what you decided in '2' above).

4) Once both groups have created their illustrations and explanations switch scenarios.

5) Once you have received the scenarios from the other half of your group figure out:
   1) Is this Penetrance or Expressivity?
   2) Which scenario illustrates and describes incomplete and which complete.

**What are the defining characteristics of Expressivity and Penetrance?**

**What in the scenarios clued you in to which (expressivity or penetrance) you were working with?**

**What will tell you if it is complete or incomplete?**

# Crossing over, recombination, and chromosome mapping

## Part 2

What is the genotype of the gametes this individual could produce?

What traits would be carried by the gametes this individual could produce?

Create a single chiasma between these homologs that crosses between two of the represented genes.

What pieces of the homologs are part of the chiasma?

What pieces of the homologs are not part of the chiasma?

Carry out crossing over and recombination by cutting your pipe cleaners and either taping or twisting the recombined sections together.

What is the genotype of the gametes this individual could produce?

What traits would be carried by the gametes this individual could produce?

Are some of the gametes after crossing over identical to those before crossing over?

Was there more variation in the gametes produces before or after you crossed over?

Note: gametes that do not undergo crossing over and recombination are called 'parental' because they carry a genotype identical to that of the parents (remember we are creating eggs and sperm here).

Crossing over, recombination, and chromosome mapping

## Part 3

Create a second chiasma between these homologs (keep the first, you are adding a second) that crosses the remaining two genes.

What is the genotype of the gametes this individual could produce?

What traits would be carried by the gametes this individual could produce?

Is there more, less, or the same variation as in the single crossover your carried out in Part 2?

If you haven't already, cut the centromeres holding the sister chromatids together.

How many crossover gametes are produced?

How many parental gametes are produced?

Note: there will always be two non-cross over gametes produced. So the maximum crossover between any two genes, ever, is 50%.

Map units (mus) are a measurement of the % crossing over between two genes. If crossing over occurs between two genes 17% of the time those genes are said to be 17 mus apart on the chromosome.

Therefore, when mapping a chromosome, it's maximum length is 50 mus (because there will always be another 50 mus that never cross over).

Crossing over, recombination, and chromosome mapping

## Part 4

Using what you learned about map units in part 3, assign map units between the genes you created on your homologs. (how far apart is each gene from the other?). *Hint: how frequently do you expect recombination to occur between each set of genes?*

**Mathematically, how would you expect the distances between the central gene and each of the furthest genes to compare to the distance between the furthest genes?**
*Hint: If you drove from Stillwater to Guthrie and stopped, then continued to Oklahoma City, would that distance be different than if you drove from Stillwater to Oklahoma City (through Guthrie) without stopping?*

Choose one of your double cross-over gametes and mate it with a homozygous recessive gamete.

**What genotype does the resulting offspring have?**

**What traits does this offspring possess (this is called its phenotype).**

Using the blank paper, markers, and colored pencils provided draw your offspring. Underneath the drawing write its genotype.

If you have time left, cross a second gamete with a homozygous recessive individual. Draw this sibling & provide their genotype as well.

Repeat with all of the gametes produced as time allows.